4 **(a)** Define gravitational field strength at a point.

.....

.....

..... [1]

(b) State the relationship between gravitational field strength and gravitational potential at a point.

.....

.....

..... [1]

(c) A satellite of mass 2200 kg at position P orbits the Earth in a circular path of orbital radius 6.70×10^6 m. as shown in Fig. 4.1.

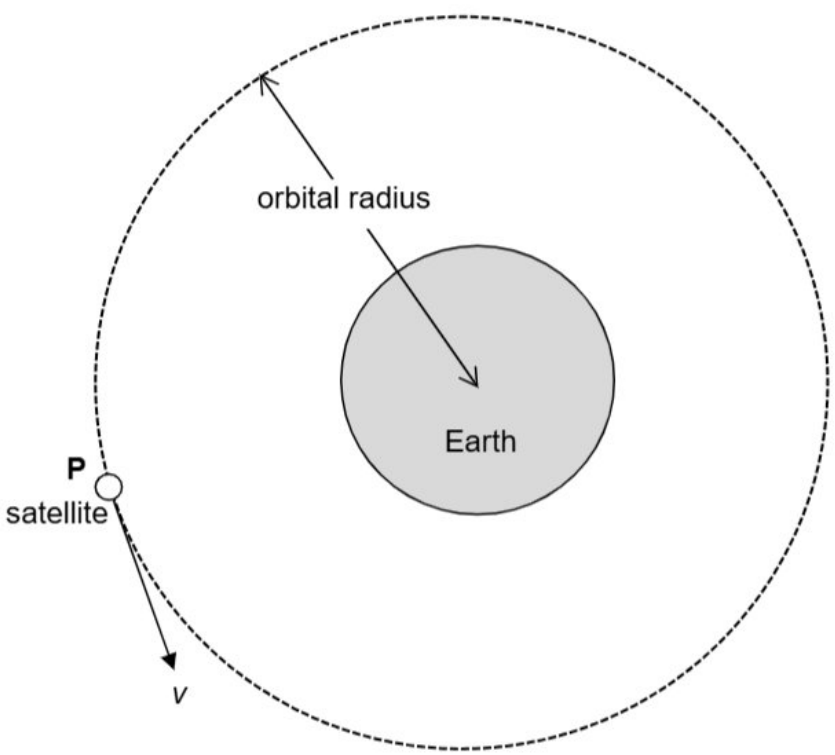


Fig. 4.1

The mass of the Earth is 5.98×10^{24} kg and the radius of the Earth is 6.40×10^6 m.

At point P,

(i) calculate the magnitude of the gravitational field strength g_P,

g_P = ... N kg^{-1} [2]

(ii) calculate the magnitude of the gravitational force on the satellite,

force = ... N [1]

(iii) state the rate of change of potential energy with orbital radius for the satellite,

rate of change of potential energy = ... J m^{-1} [1]

(iv) determine the total energy E_T of the satellite.

E_T = ... J [3]